

Kaloyan (Kal) Draganov

Boston, MA | kaloyan@draganov.com | +1 857-361-2905 | January – June 2027 Co-op Availability
<https://linkedin.com/in/kaloyan-draganov-081567155/> | <https://kkdraganov.github.io> | <https://github.com/kkdraganov>

EDUCATION

Northeastern University, Boston, MA

Expected May 2028

Bachelor of Science in Computer Engineering and Computer Science

GPA 3.9/4.0 (Dean's List)

Activities: Northeastern Mars Rover Team, AerospaceNU Avionics, Wireless Club, Cooking Club

Notable Coursework: Embedded + Mobile Robotics, Computer Systems, Electronics, OOD, Networks, Algorithms, Circuits & Signals

TECHNICAL SKILLS

Programming: Python, C, C++, Java, C#/.NET, SQL, PyTorch, MATLAB, Git

Embedded/Robotics: ROS 2, STM32, FreeRTOS, Altium, FPGA, Linux/Ubuntu, I²C, SPI, UART, Oscilloscope, Logic Analyzer, SMD soldering

WORK EXPERIENCE

Re:Build Fikst

January 2026 – June 2026

Embedded Systems Co-op

Wilmington, MA

- Developed embedded firmware in C for STM32 with FreeRTOS, implementing device control and system logic alongside TouchGFX GUI workflows and support for new PCB revisions and peripheral variants
- Performed hardware bring-up for the device main PCB using an oscilloscope and logic analyzer, developing I²C drivers for the onboard EEPROM and voltage-supervisor ICs and identifying a PCB layout issue that prompted a board respin
- Owned two projects developing production systems to automate procedures in medical device manufacturing
 - Cassette Flasher: Designed and built a Python-based standalone EEPROM programming station on Ubuntu using FT232H with multi-threaded hardware control and system deployment; used for every manufactured consumable
 - Main PCB End-of-Line QC: Led development of a fault-tolerant automated PCB test station implementing custom Windows/Linux DAQ drivers with hardware timeout handling and safe shutdown; validates every production PCB
- Unified two core components of the company's embedded development infrastructure by implementing support for its proprietary serial protocol within the embedded prototyping framework, resolving a longstanding interoperability gap

Uru Intelligence, LLC

April 2025 – September 2025

Founder & CTO

Wellesley, MA

- Architected technical stack from the ground up: React/FastAPI/Supabase app in Docker Compose hosted on Elestio
- Engineered an MCP-driven integration layer to extend standard LLMs to utilize third-party apps (e.g. Gmail, Calendar, Teams)
- Designed AI pipelines leveraging n8n and OpenAI agents, custom RAG vector databases in Pinecone, and OpenAI text embeddings for intelligent document retrieval and analysis

Dynamo Software

January 2022 – June 2024

Developer

Watertown, MA

- Independently scoped, designed, and delivered core product features in Python, SQL, and C# after collaborating with clients to define key use cases and translate business needs into technical specifications
- Architected data pipelines and backend services integrating SQL Server and RESTful APIs for automated data processing
- Designed a "signature parser" microservice in C#, creating custom pattern recognition and similarity algorithms, achieving 90% accuracy in extracting contact data from client emails and reducing manual processing time by 95%

TECHNICAL CLUBS & PROJECTS

Northeastern Mars Rover Team

March 2025 – Present

Autonomous Software Lead

Placed 3rd URC '25, 6th URC '26, 2nd CIRC '26

- Implemented EKF sensor fusion in ROS 2 on a Jetson Orin, fusing GNSS and frame-transformed ZED IMU odometry; developed noisy-sensor simulation and ground-truth visualization for validation
- Developed a holonomic MPPI local planner for obstacle avoidance with stochastic control sampling and tuned cost functions
- Migrated drivetrain motor control from PWM to CAN, integrating Kraken X44 motors on a 1 MBd CAN 2.0 bus and developing ROS 2/SocketCAN motor control with 18% measured bus utilization

AerospaceNU Avionics

September 2025 – December 2025

Embedded Engineer

- Developed flight control and telemetry software in PlatformIO C++ on an ARM Cortex-M0 processor, implementing real-time control logic for the flight computer, telemetry streams, engine ignition, and launch sequencing
- Integrated custom PCBs with onboard sensors and peripherals over SPI, I²C, and UART, ensuring synchronized operation

Spiderbot**May 2025 – June 2025**

- Developed FPGA control logic on an Intel DE10-Nano SoC (FPGA + dual-core ARM Cortex-A9 HPS) using Quartus Prime and Platform Designer implementing arithmetic, multiplexing, and timing circuits to create finite state machines
- Integrated custom Verilog FPGA modules with the ARM HPS via the HPS-to-FPGA bridge, enabling MMIO communication with a custom developed Linux C++ application for autonomous hexapod sensor reading and PWM-based servo control

Escape Room**February 2025 – April 2025**

- Engineered an RFID-based locking mechanism on Arduino, communicating with three MFRC522 readers over SPI
- Designed puzzle components in SOLIDWORKS and fabricated them with laser cutting and PLA 3D printing
- Coordinated puzzle integration across 13 teams on the Flow Committee, designing a master flow chart for a cohesive game

FIRST Robotics (Team 8567)**September 2020 – July 2024***Technical Captain*

- Directed a 50-member team in the end-to-end design, fabrication, wiring, and programming of a competitive 30" x 30" x 36" robot, qualifying for the New England District Championships and in our first season winning the Rookie All-Star Award
- Engineered the robot's electrical architecture, using CAN and PWM to control pneumatics and brushless motors
- Developed Java-based real-time control software, applying kinematics, PID feedback loops, odometry, and sensor fusion of IMU odometry with camera-based AprilTag detection to enable field-relative localization and autonomous path planning

INTERESTS

Robotics, Arduino, Beekeeping, 3D Printing, Computer Vision, Soccer, Rock Climbing, Skiing, Chess, Reading